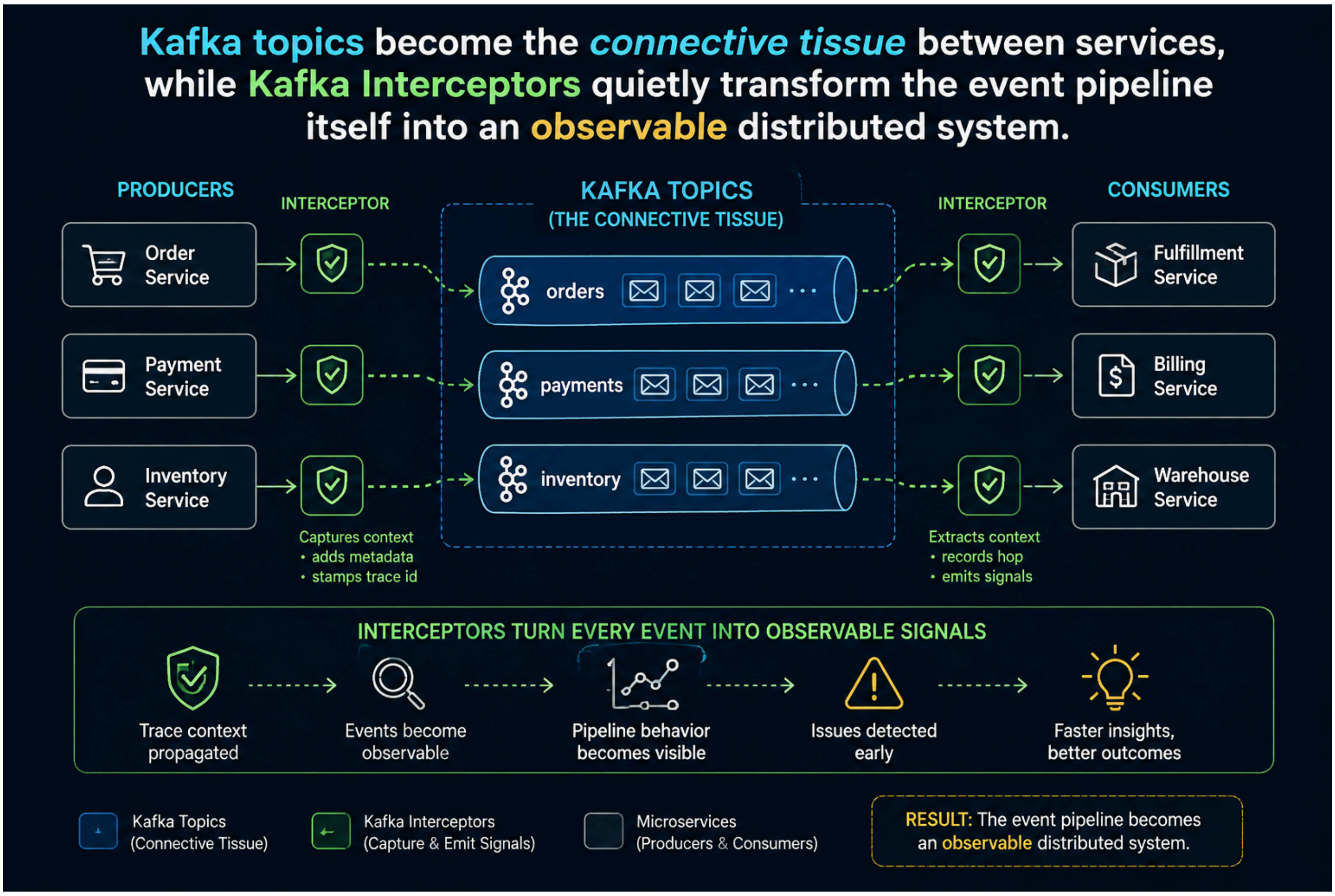
Confluent Kafka Isotope

`confluent-kafka-isotope` is a reference implementation of an ***e-commerce order pipeline that uses Kafka Interceptors, Prometheus, and Apache Flink to capture, analyze, and report end-to-end event-tracing data in both batch and near real time***.

Much like isotopes used to trace molecules through a biochemical pathway, each event carries lightweight metadata that allows it to be followed as it travels through Kafka topics and distributed microservices.



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1.0 How an Isotope Traverses an Event Pipeline

An **Isotope** is a ***lightweight tracing artifact attached to Kafka record headers***. Like a biochemical isotope used to trace molecules through a metabolic pathway, it allows the journey of a record through an event-driven architecture to be observed and analyzed.

This project models a Kafka pipeline as a **bipartite graph**: *services occupy one vertex set*, *topics the other*, and *every produce and consume operation forms an edge between them*. The resulting graph provides a unified view of the complete event topology, from producers to intermediate processing services to terminal consumers.

A `ProducerInterceptor` stamps each record with an isotope and appends one hop for every `send()` operation, capturing the produce edges. Consumers call `IsotopeContext.recordConsume()` to emit a lightweight marker representing the corresponding consume edges, while services that consume and then produce invoke `IsotopeContext.adoptFromRecord()` so the trace identity persists across every hop. Apache Flink reconstructs the complete **service → topic → service** graph from the isotope headers alone, producing seven reports: end-to-end latency, latency percentiles, produce-side topology, the full bipartite topology, hop distribution, per-topic coverage (a trace-loss funnel signal), and stuck-trace detection. The same implementation runs unchanged on both Confluent Platform (self-managed) with Confluent Manager for Apache Flink (CMF) and Confluent Cloud for Apache Flink (CCAF).

> **Full architectural design of Isotope Tracing** — the header layout (`x-isotope` JSON + seven scalar headers with a worked example), how the producer interceptor gets invoked, why the consume side uses explicit calls instead of a `ConsumerInterceptor`, and the bipartite-graph rationale — are documented in **docs/design.md**.

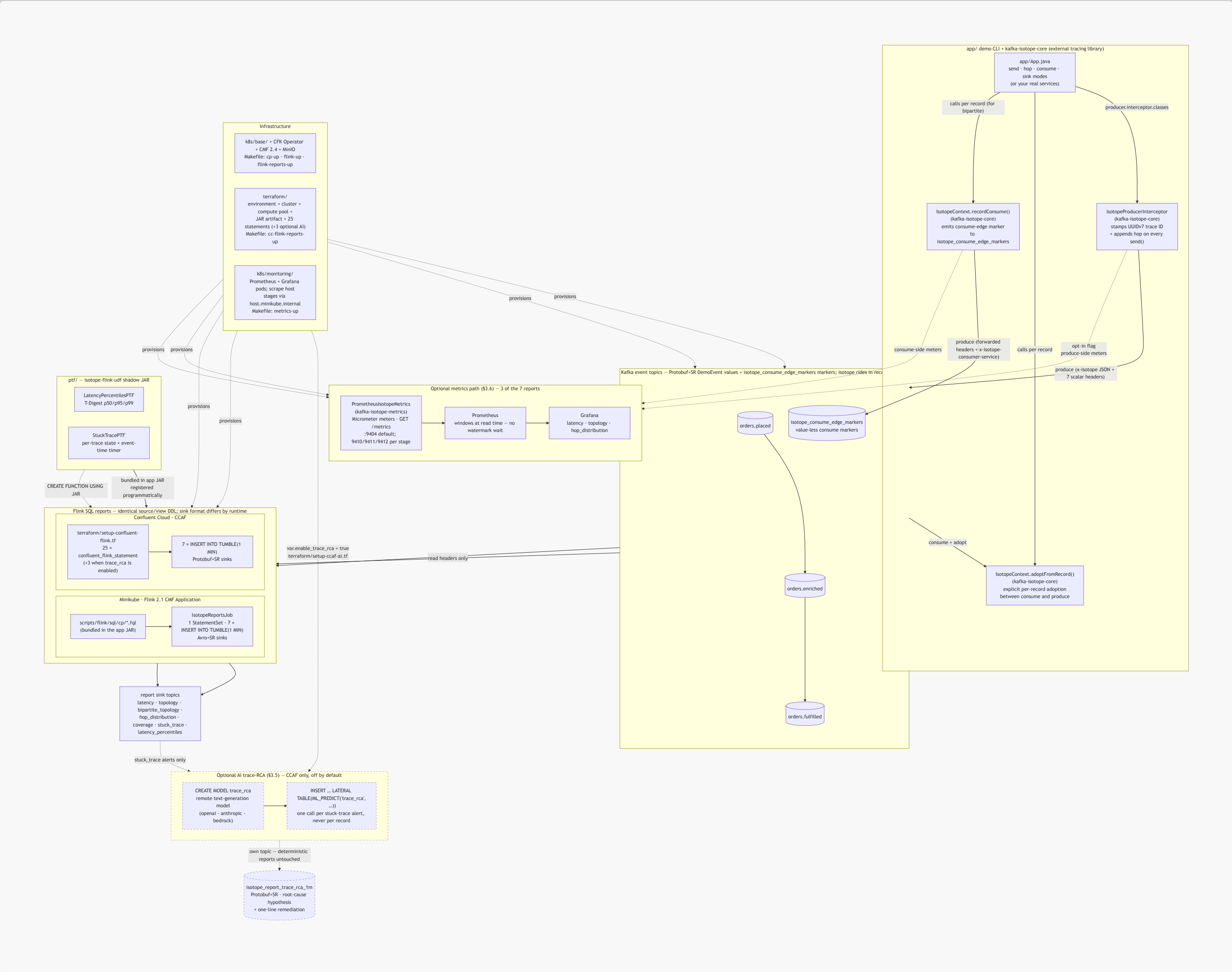
2.0 Project's Architecture

A bird's-eye view of the moving parts. The demo CLI in `app/` consumes the external tracing library (`ai.signalroom:kafka-isotope-core`), which registers a Kafka `ProducerInterceptor` that stamps an isotope into record headers on every `send()`. Consume-then-produce services propagate the inbound trace by explicitly calling `IsotopeContext.adoptFromRecord(record)`. Business events then flow through a three-topic Kafka pipeline, where Flink SQL reads the isotope metadata and emits one-minute aggregate reports.

Both runtimes run the same seven logical reports off the same source and view definitions, though each runtime keeps its own copy of the SQL. **Confluent Platform (CP)** on Minikube executes the `.fql` files under `scripts/flink/sql/cp/` as a single Flink 2.1 Confluent Manager for Apache Flink (CMF) Application (`IsotopeReportsJob`), while **Confluent Cloud for Apache Flink (CCAF)** applies the same logical SQL as inline `confluent_flink_statement` Terraform resources under `terraform/`. The shadow JAR from `ptf/`—which powers two of the seven reports—runs unchanged on both runtimes: bundled into the CP application JAR and uploaded as a Flink artifact on CCAF.

Alongside that—**additive, opt-in, and disabled by default**—the interceptor can also emit **Micrometer** metrics for **Prometheus**, with **Grafana** providing visualization (§3.6). This path produces the three **stateless** reports (`latency_1m`, `topology_1m`, and `hop_distribution_1m`) without requiring a stream processor. The remaining four reports continue to run in Flink because they depend on per-`trace_id` state or absence-of-event analysis, which falls outside Prometheus's query model.

(Kafka is drawn once below for brevity; each runtime provisions its own Kafka cluster.)



▶ See Repo Layout

3.0 Getting Started

First, clone the repo to your local machine using the [GitHub CLI](https://cli.github.com/):

```shell
gh repo clone j3-signalroom/confluent-kafka-isotope
```

Change to the repo directory:

```shell
cd /path/to/confluent-kafka-isotope
```

Then decide what you want to run:

3.1 Unit tests (no broker, instant)

```
./gradlew test                     # runs :ptf:test (the app has no unit tests in this repo)
# or scoped:
./gradlew :ptf:test                # TDigestsTest — T-Digest sketch (de)serialization + accuracy
```

3.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has two argument styles — **pipeline-position verbs** that bake in the orders.* topic chain (recommended for the demo) and **generic verbs** that take raw topic + service args (for ad-hoc inspection on any topic):

Verb	Args	What it does
`place`	`[payload]`	Produces one isotope-tagged `DemoEvent` to `orders.placed` as `order-intake-service` (default payload: `hello`), then exits. Auto-creates the topic.
`enrich`	—	Consumes from `orders.placed`, adopts the isotope, **emits a consume-edge marker to `isotope_consume_edge_markers` as `order-enrichment-service`**, then re-produces to `orders.enriched`. Runs until Ctrl-C.
`fulfill`	—	Same as `enrich` but for `orders.enriched → orders.fulfilled` as `order-fulfillment-service`. Runs until Ctrl-C.
`ship`	—	Terminal consumer for `orders.fulfilled` as `shipping-notification-service`. **Emits a consume-edge marker** so it shows up in the bipartite report; does not re-produce. Runs until Ctrl-C.
`send`	`<topic> <service> <payload>`	Generic produce. Auto-creates the topic.
`hop`	`<in-topic> <out-topic> <service>`	Generic consume-then-produce; emits a consume-edge marker. Runs until Ctrl-C.
`consume`	`<topic> <service>`	Generic terminal-consume; emits a consume-edge marker and pretty-prints the trail. Runs until Ctrl-C.
`sink`	`<topic>`	Passive peek — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection. Runs until Ctrl-C.

3.2.1 Full Bipartite Demo

A 4-stage chain (full bipartite graph) in four terminals. Run them in pipeline order — `place` produces, then `enrich` / `fulfill` / `ship` each pick up where the previous stage left off:

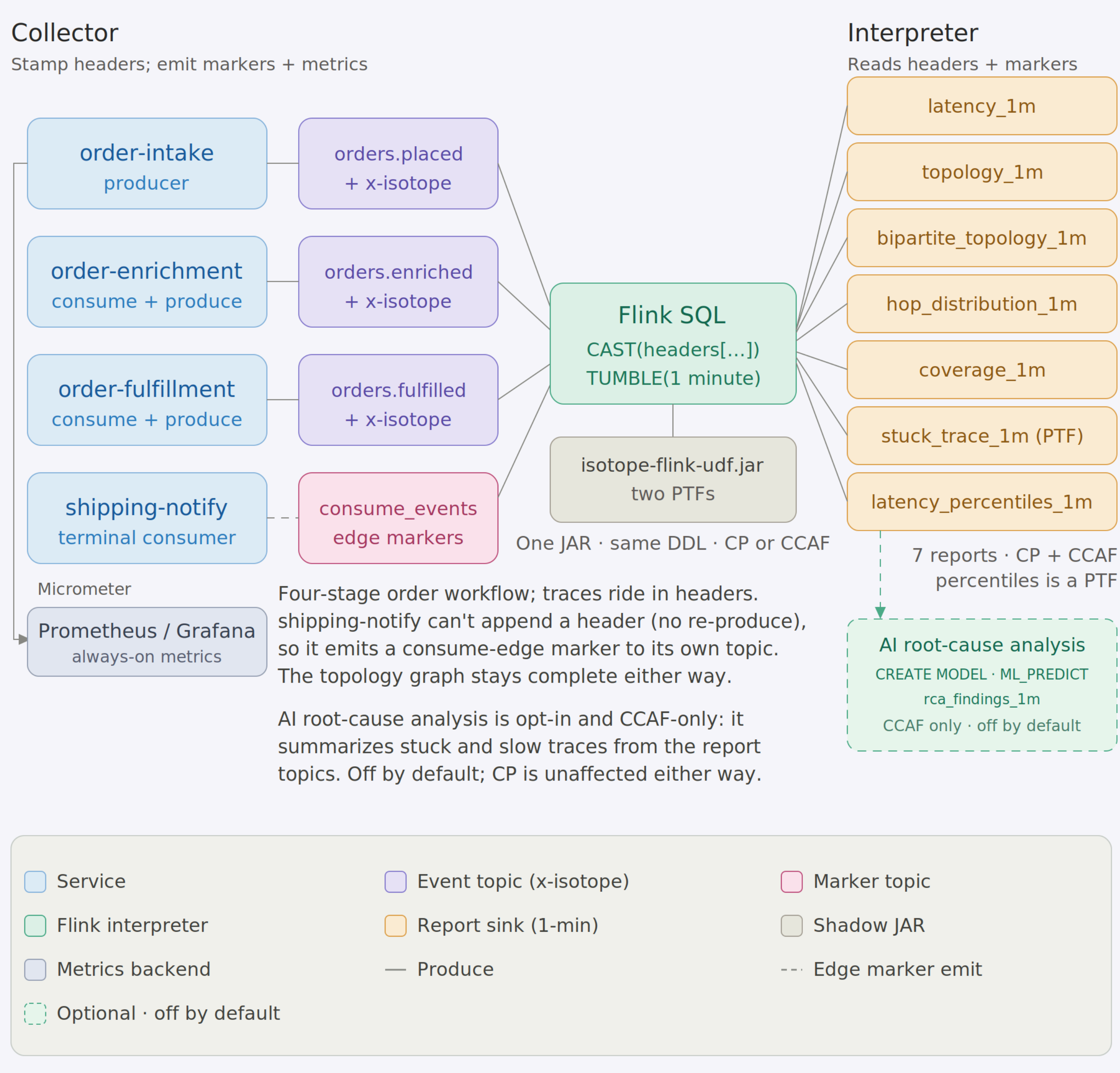
```
# Terminal A — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="place 'hello world'" -q

# Terminal B — first hop: orders.placed → orders.enriched
./gradlew :app:run --args="enrich" -q

# Terminal C — middle hop: orders.enriched → orders.fulfilled
./gradlew :app:run --args="fulfill" -q

# Terminal D — terminal consumer (prints the full 3-hop trail AND emits a
#              consume-edge marker so shipping-notification-service shows up
#              in the bipartite report)
./gradlew :app:run --args="ship" -q
```

Terminal D's output for each record shows the same `trace_id` across all three hops, `origin = order-intake-service` (never reassigned), and `hops[]` listing `order-intake-service → order-enrichment-service → order-fulfillment-service` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `order-intake-service → orders.placed`, `order-enrichment-service → orders.enriched`, `order-fulfillment-service → orders.fulfilled` and consume edges `orders.placed → order-enrichment-service`, `orders.enriched → order-fulfillment-service`, `orders.fulfilled → shipping-notification-service`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=…` / `-Dschema.registry.url=…` if you're not on the default Minikube layout.



> **Just want the commands?** The full local stack-up sequence (cluster → Kafka → Flink → reports → traffic → teardown) is consolidated in **docs/runbook-minikube.md**.

3.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start                 # one-time
make cp-up                          # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up                    # localhost:30092 → Kafka,
localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest                                          # every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'         # just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
    -PkafkaBootstrap=localhost:30092 \
    -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
`BrokerSmokeIT`	AdminClient can create/list/delete a topic via the NodePort port-forward
`ProducerInterceptorIT`	A consumer sees the `x-isotope` JSON header + all 7 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves `DemoEvent.source` / `payload`
`ThreeStageHopPropagationIT`	`order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service` produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = `order-intake-service`, this = `order-enrichment-service`, hop count = 2) at the terminal; consume-then-produce hops use `IsotopeContext.adoptFromRecord` to carry the trace forward
`BipartiteTopologyIT`	The 4-stage `order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service → topic-CD → shipping-notification-service` chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded `x-isotope-*` scalars describing the upstream producer, and the new `x-isotope-consumer-service` naming the downstream consumer. Asserts the `(consumer_service, consumed_topic)` set is exactly the three pairs of stages 2-4

3.4 Confluent Platform on Minikube — Production-Like Streaming, Running Locally

> **Caveat:** Minikube is a single-node cluster. It is not a production-like environment, but it is sufficient for local development and testing. The Confluent Platform (CP) stack runs in Minikube, and you can port-forward Kafka and Schema Registry to your local machine.

To ***run***, ***test***, and ***debug*** Apache Flink like a production engineer, this project provides a full Confluent Platform stack running locally on Minikube — no cloud required.

You get a production-like environment on your machine, with all the components you'd expect in a real deployment:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.2** via the Confluent Flink Kubernetes Operator 1.140.1
- **Confluent Manager for Apache Flink (CMF) 2.4.0** for Flink environment management

To run this project, you'll need **macOS (with Homebrew)** or **Linux (with apt-get)**. The full stack — **Minikube + Confluent Platform + Flink + CMF** — is resource-intensive and designed to mirror a production environment. Therefore, the following defaults are recommended:

Resource	Default
CPUs	6
Memory	20 GB
Disk	50 GB

> These settings ensure stable performance across all components. You can tune them as needed, but lower resource levels may cause pod restarts or degraded performance.

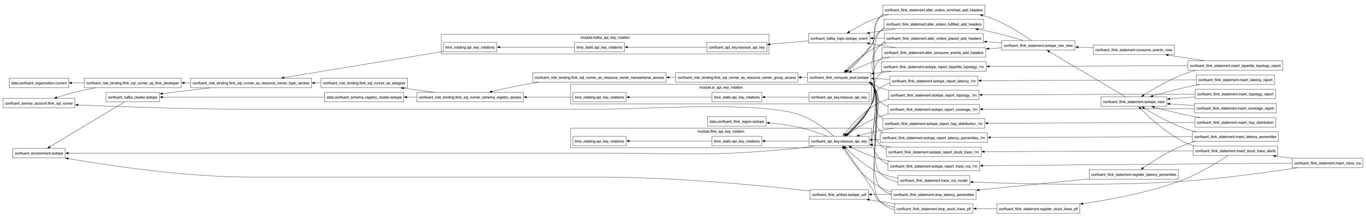
Seven reports — five pure Flink SQL plus two JAR-backed PTFs — run as a **single Flink 2.1 CMF Application** (`IsotopeReportsJob`, one StatementSet, seven sinks) submitted through CMF and executed by the Confluent Flink Kubernetes Operator. No raw session cluster is involved; `k8s/base/flink-cluster-deployment.yaml` (`make flink-deploy` / `make flink-sql`) is a separate, optional path for ad-hoc SQL. Confluent Cloud runs the same seven reports from its own copy of the SQL, inlined in Terraform — see **§3.5** for that path; this section is the local-Minikube one.

The full bring-up sequence — cluster → Flink → reports → traffic → teardown — is consolidated in docs/runbook-minikube.md. The short version: `make flink-up` then `make flink-reports-up`, then drive traffic across **multiple** 1-minute windows (a single burst sits in one open window forever — the watermark has to cross `window_end` for a tumbling window to emit) and wait ~90s after the last record.

Report sink topics ride **Avro+SR** (`avro-confluent`, auto-registered on first write) so Control Center renders them natively — a deliberate *format-by-domain* split: app events are **Protobuf+SR** (`DemoEvent`), Flink aggregates are **Avro+SR** (cp-flink ships no SR-integrated Protobuf format), and the consume-edge marker topic `isotope_consume_edge_markers` is **null-value / headers-only**. Not a defect — a clean split by domain.

3.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

The CCAF parallel of §3.4, driven by Terraform under terraform/ — no local cluster. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, compute pool, two rotating service-account API key pairs (Kafka + Schema Registry), the uploaded PTF JAR, and 25 `confluent_flink_statement` resources (4 ALTER TABLE + 3 VIEW + 7 sink CREATE TABLE + 2 CREATE FUNCTION + 7 INSERT INTO, plus 2 transient DROP FUNCTION). The shape mirrors `apache_flink-kickstarter-ii` — same provider version and `iac-confluent-api_key_rotation-tf_module`.



The full provision → deploy → traffic → teardown sequence is consolidated in docs/runbook-ccaf.md. The short version: `export` a Cloud-level Confluent Cloud API key, `make cc-flink-reports-up` (~6–8 min first run; idempotent re-applies), drive traffic with `scripts/cc-app-run.sh place|enrich|fulfill|ship` across **multiple** 1-minute windows (a single burst sits in one open window forever, and you wait ~90s after the last record), then `make cc-flink-reports-down` to `terraform destroy` the whole environment.

Format-by-runtime (not-by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in scripts/flink/sql/cp/05_report_sinks.fql). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in terraform/setup-confluent-flink.tf). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in scripts/flink/sql/cp/, CCAF's lives inline as `confluent_flink_statement` resources in terraform/setup-confluent-flink.tf.

Why percentiles is a PTF. CCAF rejects all `CREATE FUNCTION` statements for user-defined *aggregate* functions ("aggregate functions are not supported"). Percentiles would naturally be an aggregate, so to keep the report portable it's implemented as a `ProcessTableFunction` instead — `LATENCY_PERCENTILES` (class `LatencyPercentilesPTF`) does its own 1-minute tumbling-window aggregation over a T-Digest sketch via per-window state and event-time timers. A PTF has no such restriction, so it registers and runs on **both** runtimes, exactly like `STUCK_TRACE_PTF`. Both runtimes therefore run the same seven reports: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`, and `latency_percentiles` (p50/p95/p99).

Optional: AI root-cause analysis (CCAF-only, off by default). Beyond the seven deterministic reports, terraform/setup-ccaf-ai.tf wires an **eighth, AI-generated report** that turns each stuck-trace *alert* into a natural-language root-cause hypothesis plus a one-line remediation. It adds three `confluent_flink_statement` resources — `CREATE MODEL trace_rca` (a remote text-generation model), a `proto-registry` sink `isotope_report_trace_rca_1m`, and an `INSERT … SELECT … LATERAL TABLE(ML_PREDICT('trace_rca', …))` — all **gated on `var.enable_trace_rca` (default `false`)**, so a normal `make cc-flink-reports-up` is completely un``affected. The model is invoked once per *alert* (the low-volume 1-minute stuck-trace report topic), never per record, and its output lands on its own topic — it never overwrites the deterministic reports. Defaults target OpenAI (`gpt-4o`); for Claude hosted by Anthropic — a **direct** CCAF provider — set `rca_model_provider = "anthropic"`, `rca_model_endpoint = "https://api.anthropic.com/v1/messages"`, a Claude `rca_model_version`, and your Claude `rca_model_api_key` (bare API key, no AWS auth), plus the

Anthropic-required `rca_model_max_tokens`. Claude via AWS Bedrock (`rca_model_provider = "bedrock"`) also works but isn't required. The standalone SQL PoC — a `CREATE MODEL` + `ML_PREDICT` walkthrough with provider notes and alternative options — is in scripts/flink/sql/ccaf-ai/trace_rca.fql.

3.6 Stateless reports via Micrometer → Prometheus/Grafana (optional)

Three of the seven reports — `latency_1m`, `topology_1m`, `hop_distribution_1m` — are pure stateless scalar aggregation over bounded-cardinality dimensions (service / topic / hop_count, never `trace_id`). Those don't need a stream processor: the producer interceptor already has every value in scope on each `send()`, so it emits them as **Micrometer** meters (PrometheusIsotopeMetrics.java) and lets **Prometheus** window them at read time, with **Grafana** on top. The other four (`latency_percentiles`, `coverage`, `bipartite_topology`, `stuck_trace`) are per-`trace_id` stateful or absence-of-event problems Prometheus can't express, so they **stay in Flink**. This path is **additive and opt-in** — a 3-Micrometer / 4-Flink split.

> **Full details** — the meter/PromQL reference, the produce- and consume-side signals, the two deliberate gaps (`distinct_traces`, windowed `min`), and why percentiles stays a PTF — are in **docs/metrics.md**. The one-command Prometheus + Grafana showcase has its own runbook: **k8s/monitoring/README.md** (`make metrics-up`).

4.0 Resources

- Medium Article: Kafka's quiet observability superpower — Kafka Interceptors